MECH 325 - Midterm Book 1
Closed Book

Thursday, October 23, 2025

Name: Solution

Student Number: _____

Circle your section

Section 102 (11:00 Lecture)

Section 101 (9:30 Lecture)

Signature: _____

Part 1 MC Mark _____ / 12

Part 1 SA Mark _____ / 10

Part 2 LA Mark _____ / 38

Total _____ **/ 60**

Instructions

There are two parts to this exam with different instructions for each. Please read carefully.

Part 1 – Closed-Book (22 marks)

Multiple choice (12 marks: 1 mark per question). Complete all 12 questions by marking your response on the paper.

Short answer (10 marks). Complete all 5 questions by marking your response in this exam booklet.

After you have completed Part 1, hand in your paper in order to begin the Open Book portion. You may not return to Part 1 after you hand it in.

Part 2 – Open-Book (38 marks)

Long answer (38 marks): Complete all parts to the questions for Part 2 by marking your responses on the exam paper. This portion of the exam is open-book and open-notes.

All Communication Features Must be Disabled!

Part 1A: Multiple Choice (12 Points)

Answer all 12 questions. Each is worth 1 mark. Choose the single best response – a given question may have more than one choice that is correct but marks will only be given for the *best* answer.

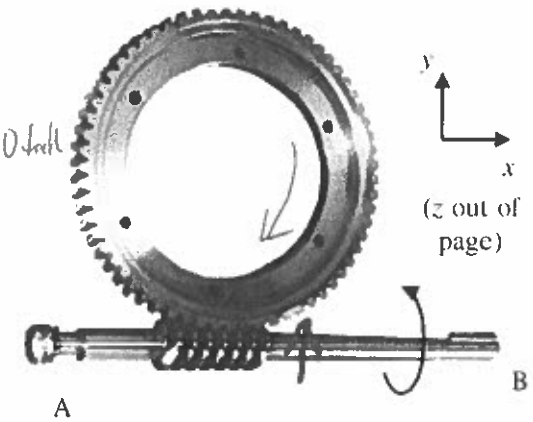
1. **What is the best description of the type of motion between the teeth of two properly meshed bevel gears?**
   a) Purely rolling
   b) Purely sliding
   c) Some rolling when teeth engage and disengage, but predominantly sliding
   d) Some sliding when teeth engage and disengage, but predominantly rolling
   e) Approximately equal parts sliding and rolling

2. **For spur and helical gears, the pressure line is tangent to which circles?**
   a) Pitch Circle
   b) Base Circle
   c) Addendum Circle
   d) Dedendum Circle
   e) Clearance Circle

3. **The worm gear shown has a 1.5" pitch diameter and has two threads. The gear has a 7" pitch diameter and 80 teeth. If the worm turns 20 revolutions in the clockwise direction indicated, how far does the gear turn?**

   $N_w = 2$
   $N_G = 80$
   $20 rev = 20 \cdot 2 = 40 teeth$
   $\therefore \frac{1}{4} rev$

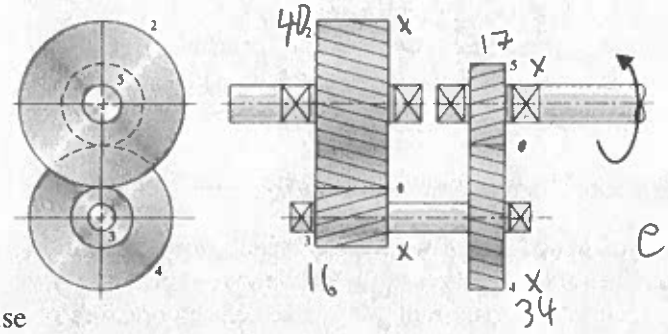
   a) ¼ rotation clockwise
   b) ¼ rotation clockwise
   c) ½ rotation clockwise
   d) ½ rotation counter-clockwise
   e) 4 rotations counter-clockwise



4. **Which of the following types of gears allows for the shafts to be offset and non-intersecting?**
   a) Bevel
   b) Spur
   c) Worm
   d) Hypoid
   e) Helical

5. **A spur gear has 40 teeth, a 14.5° pressure angle, and a pitch diameter of 2.0 in. What is the diametral pitch of the gear?**
   a) 0.05
   b) 1.42
   c) 2.86
   d) 7.0
   e) 20.0

6. **For the gear train shown, gear 5 is the driving pinion Gear 5 has 17 teeth, Gear 4 has 34 teeth. Gear 3 has 16 teeth and Gear 2 has 40 teeth. An 1800 rpm motor is driving the gear box in the Clockwise direction. What is the speed and direction of the output shaft?**



$e = \frac{17}{34} \cdot \frac{16}{40} = \frac{1}{2} \cdot \frac{2}{5}$

$e = 1/5$

$n_{out} = e\, n_{in} = \frac{1}{5}(1800) = 360$

same direction

   a) 360 rpm clockwise
   b) 360 rpm counter clockwise
   c) 720 rpm clockwise
   d) 720 rpm counter clockwise
   e) 300 rpm clockwise

7. **The motor attached to the gearset in Question 6 above is rated at 750 Watt. What is the torque on the output shaft?**

$P = T\omega \rightarrow T = P/\omega$

$T = 750 / 360\,(1/60)\,2\pi$

$T = 19.89\ N\cdot m$

   a) 13.3 N-m
   b) 19.9 N-m
   c) 26.5 N-m
   d) 35.7 N-m
   e) 53.1 N-m

8. **The term *"Pre-Extreme Horsepower"* refers to the following:**
   a) The breaking strength of a synchronous belt
   b) The power rating for a chain where roller fatigue or excessive wear are the modes of failure
   c) The power rating for a chain where link fatigue failure is the primary mode of failure
   d) The maximum horsepower rating for a chain
   e) The smallest degree of lubrication

9. **Which of the following components of belt tension present in V-Belts but normally ignored in flat belts that is the main consideration when calculating belt life?**
   a) Initial tension (F_i)
   b) Bending tension (F_b)
   c) Centrifugal tension (F_c)
   d) Tight-side tension (F_1)
   e) Slack-side tension (F_2)

10. Rank the following flexible drives from highest to lowest power capacity. Assume approximately equal dimensions and operating conditions.

a.	V-Belt	Flat belt	Roller Chain
b.	V-Belt	Roller Chain	Flat belt
c.	Flat belt	V-Belt	Roller Chain
d.	Flat belt	Roller Chain	V-Belt
e.	Roller Chain	Flat belt	V-Belt

11. For roller chains the term *chordal speed variation* refers to:

a) the speed of engagement of a chain with a sprocket
b) the relationship between the chain speed and the sprocket speed
c) the change in chain speed as the friction in the chain changes
d) the change in chain speed due to the slight rise and fall of the chain as it engages with the sprocket
e) the change in chain speed due to elongation of the chain

12. When specifying the working load of wire rope, which of the statements is FALSE:

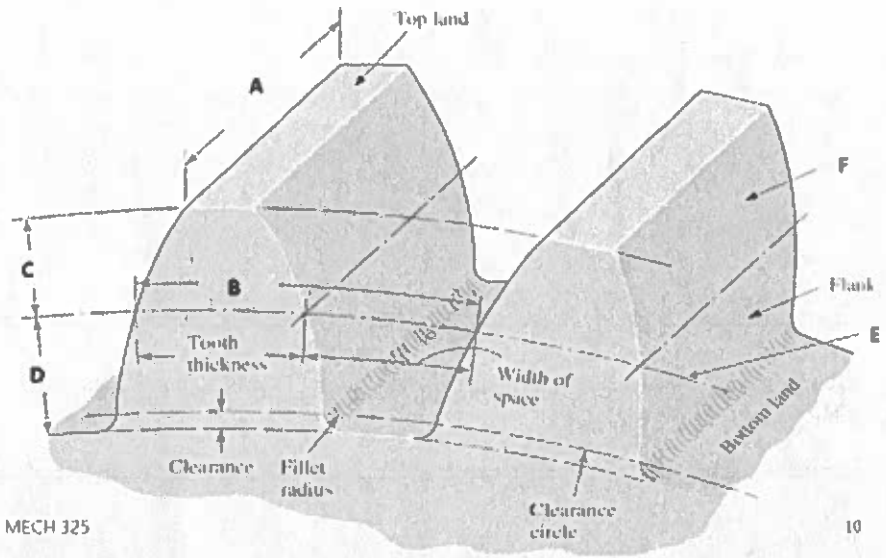
a. one must consider the modulus of elasticity of the wire rope
b. one must take in account the maximum allowable bearing pressure of the wire rope on the sheave
c. one must consider the application in order to specify a safety factor
d. one must determine the maximum sheave diameter
e. one must consider loads caused by sudden stops and starts

13. (Bonus) To convert from horsepower to ft-lbs./sec use the following conversion

a) Divide by 454
b) Divide by 33000
c) Multiply by 550
d) Divide by 550
e) Divide by 32.2

Part 1B – Short-Answer Questions (10 marks)

Answer all questions in the spaces provided.



14. Name the SIX missing terms (A through F) related to spur gears (3 marks)

A = Face Width (will accept Crown or Depth)

B = Pitch

C = Addendum

D = Dedendum

E = Pitch Circle (Pitch Diameter)

F = Face

15. Which of the above terms are important for stress analysis of the gear teeth? Why (3 marks)?

The pitch, pitch circle, and width are key terms for stress analysis.
Both contact & bending stress decrease as width increases. An increase in pitch (reduction in diametrical pitch) increases tooth size which reduces bending stress. Increasing pitch circle will reduce transmitted load (W_t), for the same torque, which will reduce the stress (bending & contact).

16. What are the advantages of chain drives over V-belts? Are there any disadvantages? (2 marks)?

Advantages: Chains are positive drive so they can absorb a high degree of load (much higher than V-belts). Chain drives require no pre-tension & can can operate at slow speeds.

Disadvantages: Chains require lubrication & have chordal speed variation.
Chains have lower efficiency than V-belts.

17. For steels, what is the difference between "through-hardening" and "case hardening?" What is meant by the "OQT1300?" (2 marks)

Through-hardening is when entire component (such as a gear) is heated above the critical temperature and then quenched in oil or water. The hard & brittle phase is then tempered to increase ductility. The entire component has the same mechanical properties (HB, S_y, S_{ut}, etc.).
Case-hardening focuses on only changing the properties of the surface of the component (such as gear teeth). This is done through concentrated heat (induction/flame) or by increasing the local carbon content through carbonizing/nitriting.

OQT 1300 is a through-hardening technique where the component is quenched (Q) in oil (O) and tempered (T) at 1300°F.

MECH 325 - Midterm Book 2
Open Book

Thursday, October 23, 2025

Name: Solution

Student Number: _____

Circle your section

Section 102 (11:00 Lecture)

Section 101 (9:30 Lecture)

I adhered to the policy of All Communication Features Disabled.

Signature: _____

Instructions

Please read carefully.

No Communication Devices Allowed!

There are TWO Problems to complete.

Part 2 – Open-Book (38 pts)

Long answer: Complete all parts to the question by marking your response in the booklet provided. This portion of the exam is open-book and open-notes. **At the end of the exam, return this handout with your answers in the booklet.**

This is a multi-part problem. For each part, use the parameter values given. If you cannot complete one part, you may assume some values and move on in order to receive marks for subsequent parts. Clearly indicate your final answer to each part by drawing a box around it. Separate different parts with a line drawn across the page.

Long Answer Problem (38 marks)

Problem 1 (16 Marks)

A V-belt system is required to attach an electric motor to a textile machine. The following information relates to the equipment

Driver	AC Motor (Normal Torque)
Driven Machine	Textile Machine
Service	10 hours/day
Input Speed	1150 rpm
Input Power	40 HP $= H_{nom}$
Output Speed (Nominal)	590 rpm
Centerline Distance (Nominal)	32 inches

a) Specify a suitable service factor
b) Specify a belt size, sheave sizes and number of belts. What is the final velocity ratio?
c) Specify the final center distance.
d) What is the safety factor for your design? Comment on the result

a) Service Factor M+H Table 7-1 DriveR: AC Motor (normal)
$K_s = 1.3$ DriveN: Textile Machinery
Time: 6-15 hrs/day
(Shigley Table 17-15 $1.2 < K_s < 1.4$) 2pts

b) Design Power $H_D = K_s H_{nom} = 1.3(40) = 52\,HP$

From Figure 7-13 $N = 1150$ rpm & $H_D = 52$ HP
Choose 5V or 5VX belts

Velocity Ratio $= VR = 1150/590 = 1.949 \sim 2.0$

Determine Sheave Sizes from Fig. 7-15 (5V belts) 1pt
Let's limit belt speed to less than 5000 fpm
Say $V = 5000$ fpm $= \frac{\pi D n}{12} = \frac{\pi D (1150)}{12} \Rightarrow D \leq 16.6''$

Options include:

D_1	D_2	VR
13.9	27.7	1.99
13.1	27.7	2.11
12.4	21.1	1.70
11.7	21.1	1.80
10.8	21.1	1.95 ←
10.2	21.1	2.06
9.65	21.1	2.18

(2pt)

5V belt

Select $D_1 = 10.8$
$D_2 = 21.1$
as best option
Final VR = 1.95

From Fig 7-15
@ 1160 rpm & 10.8" Sheave
$H_{tab} \cong 23$ HP
Since $H_D = 52$ HP

We need 3 belts

Since 1150 rpm is close to 1160 rpm use this curve on Fig. 7-15

Output Speed = $1150\text{rpm}(10.8/21.1) = \boxed{589\text{ rpm} = N_{output}}$ (1pt)

c) Specify a trial center distance CD Nom. CD = 32" (given)

$D_2 < CD < 3(D_2 + D_1)$ Eqn. 7-17

$21.1 < CD < 3(21.1 + 10.8) \Rightarrow 21.1" < CD < 95.7"$

Choose CD = 32 for this application (1pt)

Perimeter Length $L_p = 2CD + 1.57(D_2 + D_1) + \frac{(D_2 - D_1)^2}{4CD}$ (Eq'n 7-12)

$L_p = 2(32) + 1.57(21.1 + 10.8) + \frac{(21.1 - 10.8)^2}{4(32)} = 122.9$ inches

From Table 7-2 for 5V belts one can choose 118 or 125 inches

Select 125" for this application

Actual CD = $\frac{B + \sqrt{B^2 - 32(D_2 - D_1)^2}}{16}$ $B = 4L_p - 6.28(D_2 + D_1)$ (Eqn 7-13)

$B = 4(125) - 6.28(21.1 + 10.8) = 299.67$

$CD = \left[299.67 + \left[299.67^2 - 32(21.1 - 10.8)^2\right]^{1/2}\right]/16$ (2pt)

$\boxed{CD = 37.10"}$ Final Center Distance

$\boxed{\text{Belt Length} = 125"}$

Wrap Angle $\theta_1 = 180 - 2\sin^{-1}\left(\frac{D_2 - D_1}{2CD}\right)$ Eq'n 7-14

(1 pt)

$\theta_1 = 180 - 2\sin^{-1}\left(\frac{21.1 - 10.8}{2(37.10)}\right)$

$\theta_1 = 164.0°$ s. $\theta_2 = 196.0°$

Corrected Power Rating per belt = $H_{allowable} = H_a$

(1 pt)

$H_a = C_\theta C_L (H_{tab} + \Delta P)$

From Fig 7-17 $\Delta P = 1.26$ @ ~1160 rpm & VR = 1.95)

From Fig. 7-18 $C_\theta = 0.96$ @ $\theta_1 = 164°$

(3 pt)

From Fig. 7-19 $C_L = 1.0$ Belt length = 125"

From Fig 7-15 $H_{tab} = 23$ HP @ $D_1 = 10.8$" & $N_1 = 1160$ rpm

$H_a = 0.96(1.00)(23 + 1.26) = 23.29$ HP $= H_{allow}$/belt

Min. Number of Belts = $N_B > N_a > H_d$

$N_B > 23.29 > 52$

$N_B = 3$ belts

Safety Factor $n_{fs} = H_{all}/H_d = \frac{(3.0)(23.29)}{52} = \frac{69.87}{52}$

(2 pt)

$n_{fs} = 1.34$

Select Three 5V belts Length = 125 inches

$D_1 = 10.8$" $D_2 = 21.1$"

$CD = 37.10$"

Output Speed = 589 rpm

Safety Factor = 1.34

Problem 2 (22 Points)

Part 1 of 3

(7 points) Design a gear box be used to provide a required **nominal** speed of 650 rpm using full depth involute **SPUR** gears:

Note: you can choose imperial or metric units

Driver	AC Motor (Normal Torque)
Driven Machine	Liquid Pump
Service	6 hours/day
Input Speed	1750 rpm
Output Speed	650 rpm (nominal)
Input Power	10 HP (7.5 kW)
Pressure Angle	20°
Diametrical Pitch (Normal)	10 teeth/inch (module = 2.5 mm.)
Pinion	17 teeth
Gear	? teeth

Determine the following:

a) Given 17 teeth for the pinion, select a gear and determine the gear ratio (velocity ratio)
b) Pitch diameter for pinion and gear.
c) Pitch line speed
d) Tangential, radial, and normal forces for the gears

a) Gear Ratio $VR_{nom} = 1750/650 = 2.69$ $N_G = 2.69(17) = 45.76$

(1 pt) Choose $N_G = 46$ teeth so $VR = 46/17 = 2.705$

Output Speed $= 1750/2.705 = 647$ rpm

b) Pitch Diameter given $P_D = 10$ teeth/inch

(1 pt) $d_{pinion} = 17/10 = 1.700$ inches
$d_{gear} = 46/10 = 4.600$ inches

c) Pitch line speed $V = \pi \frac{d}{12} n = \pi (1.7/12) 1750$ rpm

(1 pt) $V = 778.9$ feet/minute

d) Torque = Power/ω

1pt

given $T = 63025\ HP/rpm = 63025(10/1750)$

$T = 360.1\ lb\cdot in$

$W_t = T/(d_p/2) = 360.1/(1.7/2)$

$W_t = 423.7\ lbf$

Radial load $W_r = W_t \tan\phi$ ϕ = pressure angle = 20°

$W_r = 423.7(\tan 20°) = 154.2\ lb$

2pt

$W_t = 423.7\ lbf$
$W_r = 154.2\ lbf$
$W_n = 450.9\ lbf$

$W_n = [W_t^2 + W_r^2]^{1/2}$

$= [423.7^2 + 154.2^2]^{1/2}$

$= 450.9\ lb$

Part 2 of 3

(11 Points) Determine the AGMA **contact** stress number for the 17-tooth **pinion ONLY.**

Given:

Face Width	1.5 inch (40 mm.)
Pressure Angle	20°
Gear Quality	$A_v = 8$ ($Q = 9$)
Material for Pinion & Gear	Steel
Gear Box Enclosure	Commercial Quality
Geometry factor	I= ?
Overload Factor	**$K_o = 1.25$** (uniform power source and light shock)
Dynamic factor	K_v = ?
Load Distribution factor	K_m = ? (For Shigley Text: $C_{mc} = C_{pm} = C_e = 1.0$)
Size Factor	K_s = ?
Rim thickness factor	K_B = ?
Contact stress number	s_c = ?

AGMA Contact Stress

$$s_c = \sigma_c = C_p \sqrt{\frac{W_t}{F D_p I} \cdot K_o \cdot K_r \cdot K_m \cdot K_v}$$

(M&H Eq'n 9-17, Shigley Eq'n 14-16)

Elastic Coeff. C_p: Use Table 9-7 (M&H) or Table 14-8 (Shigley)

(2pts) $\boxed{C_p = 2300 \sqrt{psi}}$ for steel on steel

Geometry Factor I

(i) M&H: Use Figure 9-17 $N_p = 17$ $\phi = 20°$ $m_G = 2.70$

$\boxed{I = 0.096}$

ii) Shigley: Use Eq'n 14-23

(2pts)

$$I = \frac{\cos\phi \sin\phi}{2 m_N} \frac{m_G}{m_G + 1}$$

$m_N = 1.0$ for Spur Gears

$m_G = 2.70$ $\phi = 20°$

$$I = \frac{\cos 20° \sin 20°}{2(1.0)} \left(\frac{2.70}{2.70+1}\right)$$

$\boxed{I = 0.117}$

$$s_{c\,min} = C_p \sqrt{\frac{W_t}{F D_p I}}$$

$$= 2300 \sqrt{\frac{423.7}{1.5(1.7)(0.096)}}$$

$\boxed{s_{c\,min} = 95.6 \text{ ksi}}$

Dynamic Factor K_v

For pitch line speed = 779 fpm ≈ 800 fpm

Gear Quality = $A_v = 8$ [Q = 9]

(2pts) From Fig 9-16 (M&H) or Fig. 14-9 (Shigley)

$K_v \doteq 1.16$

or $K_v = \left(\frac{A + \sqrt{v}}{A}\right)^B$ $\quad A = 50 + 56(1-B)$

$B = 0.25(12 - Q_v)^{2/3} = 0.25(12-9)^{2/3} = 0.520$

$A = 50 + 56(1 - 0.52) = 76.88$

$K_v = \left(\frac{76.88 + \sqrt{779}}{76.88}\right)^{0.52} = \boxed{1.17 = K_v}$

Load Dist. Factor K_m

$K_m = 1.0 + C_{pf} + C_{ma}$ (M&H Eq'n 9-17)

C_{pf} = pinion proportion factor $= \frac{F}{10 D_p} - 0.0375 + 0.0125F \quad [1.0 \le F < 15]$

(3pts) $C_{pf} = 1.5/10(1.7) - .0375 + .0125(1.5) = 0.0694$

C_{ma} = mesh alignment factor

*Assume Commercial Quality Gearbox

$C_{ma} = 0.127 + 0.0158F - 1.093\times10^{-4}F^2 = 0.127 + 0.0158(1.5) - 1.093\times10^{-4}(1.5)^2$

$C_{ma} = 0.1505$

$K_m = 1.0 + 0.0694 + 0.1505 = \boxed{1.22 = K_m}$

$K_o = 1.25$ Given

Rim $K_B = 1.0$ since solid pinion

Size $K_s = 1.0$ since $P_d > 5$

Modified $S_c = \sigma_c = 2300\sqrt{\frac{423.7}{(1.5)(1.7)(0.096)}}(1.17)(1.22)(1.25)(1.0)(1.0)$ ← M&H

(2pts) $\boxed{S_c = 127.8 \text{ ksi [M\&H]}}$

Shigley $I = 0.117$ $\quad \boxed{S_c = 115.8 \text{ ksi [Shigley]}}$

Part 3 of 3

(5 points) Select a suitable through hardened steel for the **pinion**. Specify the alloy and heat treatment.

The gear is expected to operate for the machine 200 days/year for 1 years. You can also assume the pinion and gear have the same Brinell hardness. State all assumptions.

What is your safety factor for contact stress? Comment on the result

Would you suggest flame or induction hardening for the pinion? Explain.

1pt

From Part 1 Service = 6 hrs/day

$$\#\text{cycles} = (1750\ \text{rpm})(60\ \text{min/hr})\cdot(6\ \text{hr/day})(200\ \text{day/year})(1\ \text{year})$$
$$= 1.26\times10^8\ \text{cycles}$$

1pt

From Fig 9-22 Mott or Fig 14-15 Shigley

$$Z_n = 0.95 \quad \text{or} \quad Z_n = 1.4488(1.26\cdot10^8)^{-.023} = 0.943$$

$$\text{Min } S_{ac} = \frac{S_c K_T K_R}{Z_n} \qquad K_T = 1.0 \text{ for } T < 120^\circ C \text{ assumed}$$
$$K_R = 1.0 \text{ for Reliability} = 0.99 \text{ assumed}$$

$$S_{ac} = \frac{127.8\ \text{ksi}\,(1.0)(1.0)}{0.94} = 135.5\ \text{ksi [Mott Sol'n]}$$

Choose Grade 1 Steel

$$S_{ac} = 0.322\,HB + 29.1\ \text{ksi} \Rightarrow 135.5 = 0.322\,HB + 29.1\ \text{ksi}$$

Min HB = 330 Brinell Hardness

Typical Gear Materials AISI 1045, 4140, 4340, 4150, 4750 Table 9-3

1pt

For HB > 330 one can select alloys & heat treatments such as:

4140 OQT 1000	HB = 341	17% elongation	s.f. = 34/330 = 1.04
4140 OQT 700	HB = 461	13% "	s.f. = 1.4
4340 OQT 1000	HB = 363	16% "	s.f. = 1.10
4340 OQT 700	HB = 461	12% "	s.f. = 1.40

1pt

Select 4340 OQT 1000 with suitable ductility (16%) and s.f. = 1.10

As per AGMA S.F hrl... 1.1 - 1.4

DO NOT TURN OVER